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Session : Technical Paper Presentation

MULTI-AGENTIC AI FOR NON-REVENUE WATER

Presented By

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INTRODUCTION: What is Non-Revenue Water (NRW)??

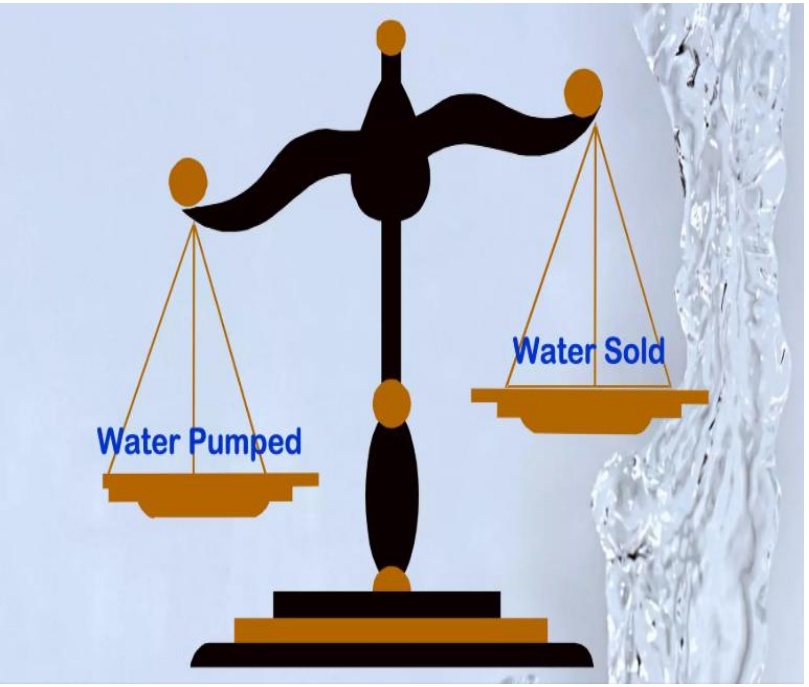
Non-Revenue Water (NRW) *is the difference between the volume of water injected into the distribution system and the volume that is billed to customers.*

Current Scenario
Utility Supply for Indian metropolitan is **500 million litres per day (MLD)**

Loss of Water
India faces loss of **40–50% of its potable water** as Non-Revenue Water—far above the global **best-practice benchmark of under 10%**

Technology
Rely on **hydraulic modelling and statistical Supervisory Control and Data Acquisition (SCADA)** systems technological interventions

Limitations
Existing systems struggle with inconsistent pressure readings in Intermittent Water Supply (IWS) networks and cannot utilize extensive unstructured "dark data" such as handwritten logs, PDF records, and multilingual customer complaints



*NRW leads to major **financial losses**, energy wastage, and increased carbon footprint for utilities*

Why Non-Revenue Water Matters

Headers can be Better
In 2 words
The Scale of the Challenge
1. Impact – not available for clean water
Kms of pipelines & how many are damaged
If detected don't for fix & still wasted
Less effort than wastage

The Scale of the Challenge

- Indian utilities often manage **2,000–8,000 km** of pipelines; **20–30%** are in poor condition.
- When leaks are detected, **30–40% remain unrepaired** for weeks, continuous water loss.
- Repairing a typical leak saves **50,000–100,000 liters/day** far more than the effort invested

Root Causes Are Complex & Multi-Layered

- **Physical losses:** Pipe bursts, aging networks, illegal digging & roadworks.
- **Commercial losses:** Meter inaccuracies, unauthorized connections, billing gaps.
- **"Dark data"** resides in PDFs, handwritten logs, and WhatsApp updates in local languages — not captured by legacy systems

Why AI-Driven Multi-Agent Systems Are Needed

- **Adaptive LLM reasoning:** Interprets natural language, manages ambiguity, and blends logs for water networks
- **Multilingual data intelligence:** Reads WhatsApp, handwritten notes, mixed languages, reduces false alarms
- **Field-to-digital insight**: Detects anomalies in noisy, intermittent systems by integrating roadwork and human context for real-time decisions

Impact: Faster, Smarter NRW Reduction

- Multi-agent reasoning reduces detection-to-repair time **from 48 hrs → 4 hrs**
- **False positives drop by 35%** when LLM agents contextualize SCADA spikes with schedule PDFs

*The rise of **Generative AI & LLM-Based Multi-Agent Systems** enables smarter, context-aware water network monitoring*

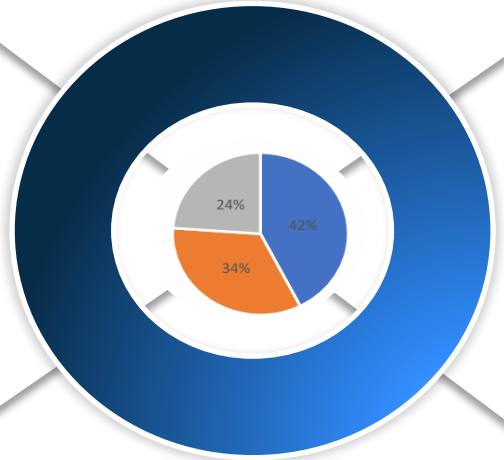
How Much Water India Loses to NRW

National NRW Level in 2025

- India's urban water utilities lose **an average of 38% of their treated potable water** to Non-Revenue Water (NRW) in 2025
- Water lost through leaks increases energy use and greenhouse gas emissions, especially in cities powered by thermal energy

Economic and Environmental Impact of NRW

- Increased dependence on costly water sources such as groundwater or distant reservoirs
- Reduced revenue for utilities, weakening capacity for infrastructure upgrades
- Greater stress on already scarce urban water resources



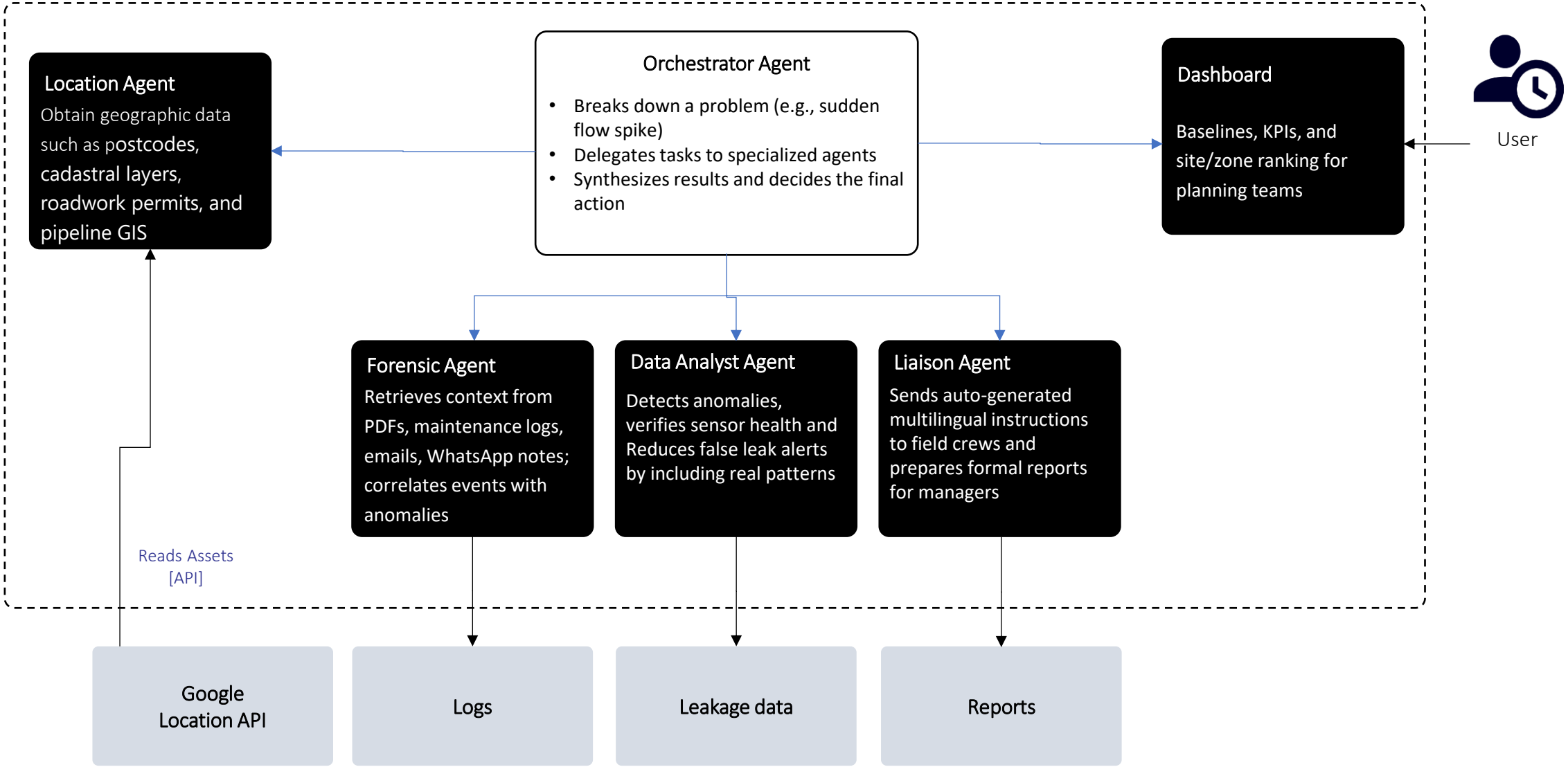
Why India's NRW Levels Are High?

- **Physical losses:** Physical loss through systemic leaks accounts
- **Apparent losses:** Theft, illegal connections, and inaccurate metering/billing
- The aging infrastructure and inadequate metering systems significantly contribute to these inefficiencies

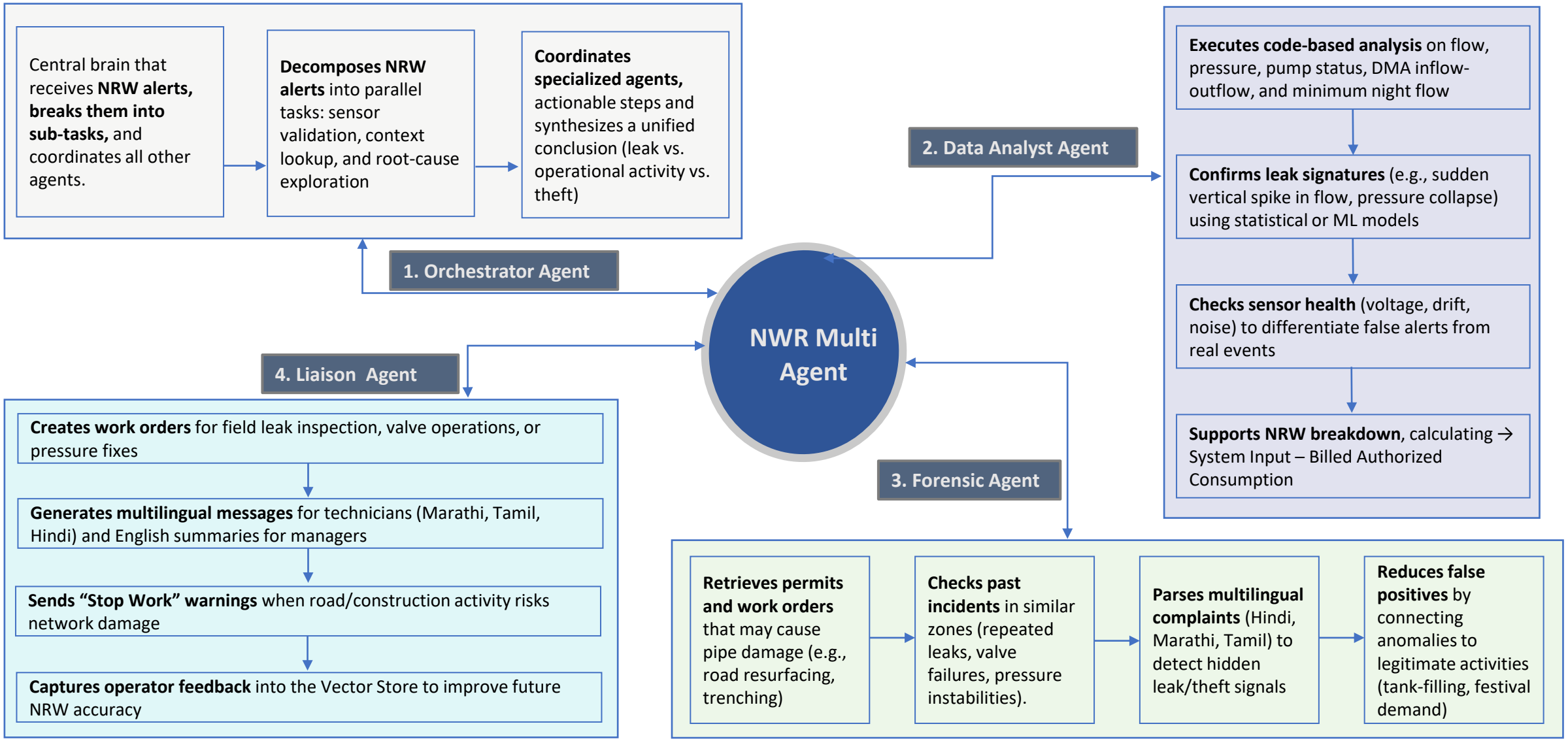
Scale of Water Lost in Major Indian Cities

- **Delhi:** ~58% NRW, failing to account for over **2,400 MLD** of its daily water supply
- **Mumbai:** ~30% NRW, losing around **1,155 MLD**
- **Chennai:** NRW reported up to **30%**

Multi Agnetic AI for Non-Revenue Water



NWR Agents and Specifications



USE CASE / CASE STUDY

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PROBLEM

- **Delhi's NRW ≈ 58%**, the highest among Indian metros.
- Delhi Jal Board (DJB) **fails to account for over 2,400 MLD** of water out of the **4,207 MLD** supplied daily. These losses include: leakages in aged pipelines
- unbilled consumption in JJ (slum) clusters
- unauthorized colonies
- illegal tapping

KPIs (what are KPIs in demo)

Cost of supplied water (production cost): = ₹0.02 per litre (₹20 per m³)

Billed Rate = ₹0.008–₹0.010 per litre (₹8–10 per m³)
Production cost of water lost:
2.4 billion litres × ₹0.02 = ₹48 crore/ day

Revenue not collected

2.4 billion litres × ₹0.0084 = ₹20.16 crore per day lost revenue

Total Daily Economic Loss (2025)

≈ ₹68+ crore per day

OUR SOLUTION

NRW Improvement	Responsible Agents	Stats
Faster Leak Detection	Data Analyst + Forensic + Orchestrator	48h → 4h; Saves 5–10 MLD/incident; ₹2–6 crore/day
Fewer False Alarms	Forensic + Orchestrator	False positives ↓ 35%; Response speed ↑ 6–12 hrs
Commercial Loss Detection	Data Analyst + Forensic + Memory	Identifies 10–30% underbilling zones; flags 8–10% JJ losses
Autonomous Workflow	Orchestrator + Liaison + GIS Forensic	Prevents 24–72 hr leak delays; recovers 50–150 MLD; saves ₹10–30 crore/day

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